

BNP to NT-proBNP Substudy Report

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1 Overview

This report compiles the figures and tables produced by the analysis script and describes how each output corresponds to the published protocol.

2 Baseline Characteristics

Baseline characteristics are summarized for the initial cohort (Table 1).

Table 1: Baseline characteristics of the initial cohort.

Characteristic	Overall (N = 460)
Age (years)	69.0 (60.0, 75.0)
Sex	
F	230 (50.0%)
M	230 (50.0%)
Body Mass Index (kg/m ²)	30.1 (25.9, 36.1)
Creatinine Clearance (mL/min)	85.6 (62.6, 113.9)
Hemoglobin (g/dL)	13.7 (12.6, 14.7)
BNP (pg/mL)	
Mean (SD)	66 (95)
Median (Q1, Q3)	36 (15, 73)
NT-proBNP (pg/mL)	
Mean (SD)	438 (957)
Median (Q1, Q3)	168 (79, 355)
Atrial Fibrillation	57 (12.4%)
Coronary Artery Disease	83 (18.0%)
Myocardial Infarction History	0 (0.0%)
Peripheral Vascular Disease	59 (12.8%)
Stroke History	22 (4.8%)
TIA History	26 (5.7%)
COPD	62 (13.5%)
Cancer History	114 (24.8%)
Congestive Heart Failure History	20 (4.3%)
Diabetes Mellitus	114 (24.8%)
Insulin Use	28 (6.1%)
Hypertension	295 (64.1%)
Obstructive Sleep Apnea	137 (29.8%)
Surgery Category	
Low Risk Surgery	79 (17.2%)
Major General	112 (24.3%)
Major Neurological	30 (6.5%)
Major Orthopedic	111 (24.1%)
Major Thoracic	16 (3.5%)
Major Uro/Gyn	43 (9.3%)
Major Vascular	44 (9.6%)
No Surgery	25 (5.4%)

¹ Number missing: CrCl (9); Hb_gdl (1); BNP (3); DM (1)

Abbreviation: Median (IQR) for most continuous; Mean (SD) & Median (IQR) for BNP & NTproBNP. Statistics calculated on non-missing data.

3 Primary Analysis: Kasahara Conversion Formula

The Kasahara formula was applied to BNP values to predict NT-proBNP. Agreement between measured and predicted values is summarized below.

Table 2: Kasahara model performance metrics (includes RMSE and R-squared).

metric	estimator	estimate
rmse	standard	727.381
rsq	standard	0.754
mae	standard	248.894
pearson_r	pearson	0.869
accuracy	binary	0.833
sens	binary	0.467
spec	binary	0.994
ppv	binary	0.970
npv	binary	0.809
kap	binary	0.539
accuracy	multiclass	0.547
sens	macro	0.463
spec	macro	0.839
ppv	macro	0.650

npv	macro	0.855
kap	multiclass	0.344
weighted_kappa_quadratic	weighted	0.656
calibration_intercept	OLS	-34.428
calibration_slope	OLS	2.395
paba_intercept	PaBa	0.710
paba_intercept_lower	PaBa	-3.131
paba_intercept_upper	PaBa	4.730
paba_slope	PaBa	0.510
paba_slope_lower	PaBa	0.468
paba_slope_upper	PaBa	0.561

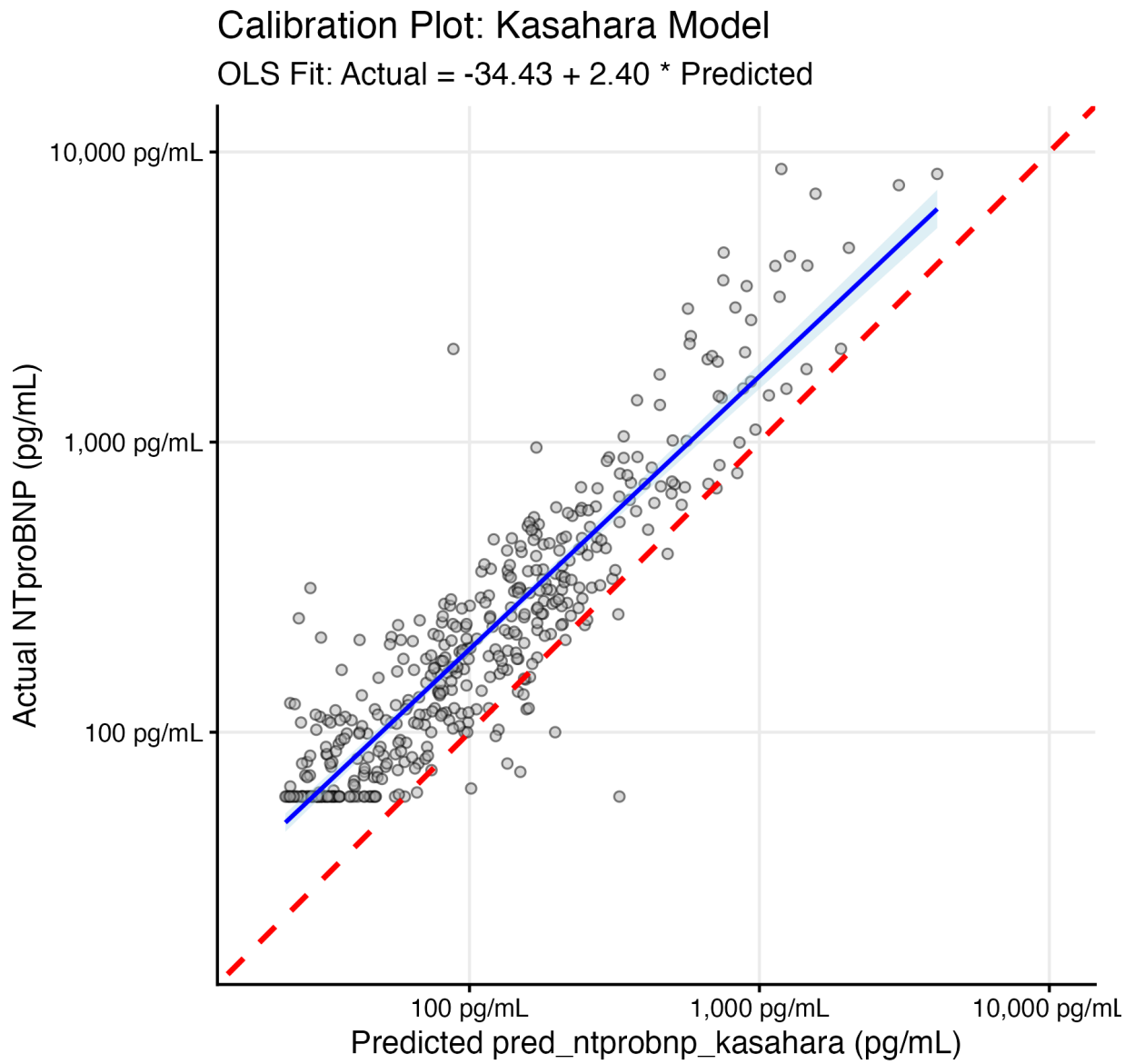


Figure 1: Kasahara model calibration plot (log scale).

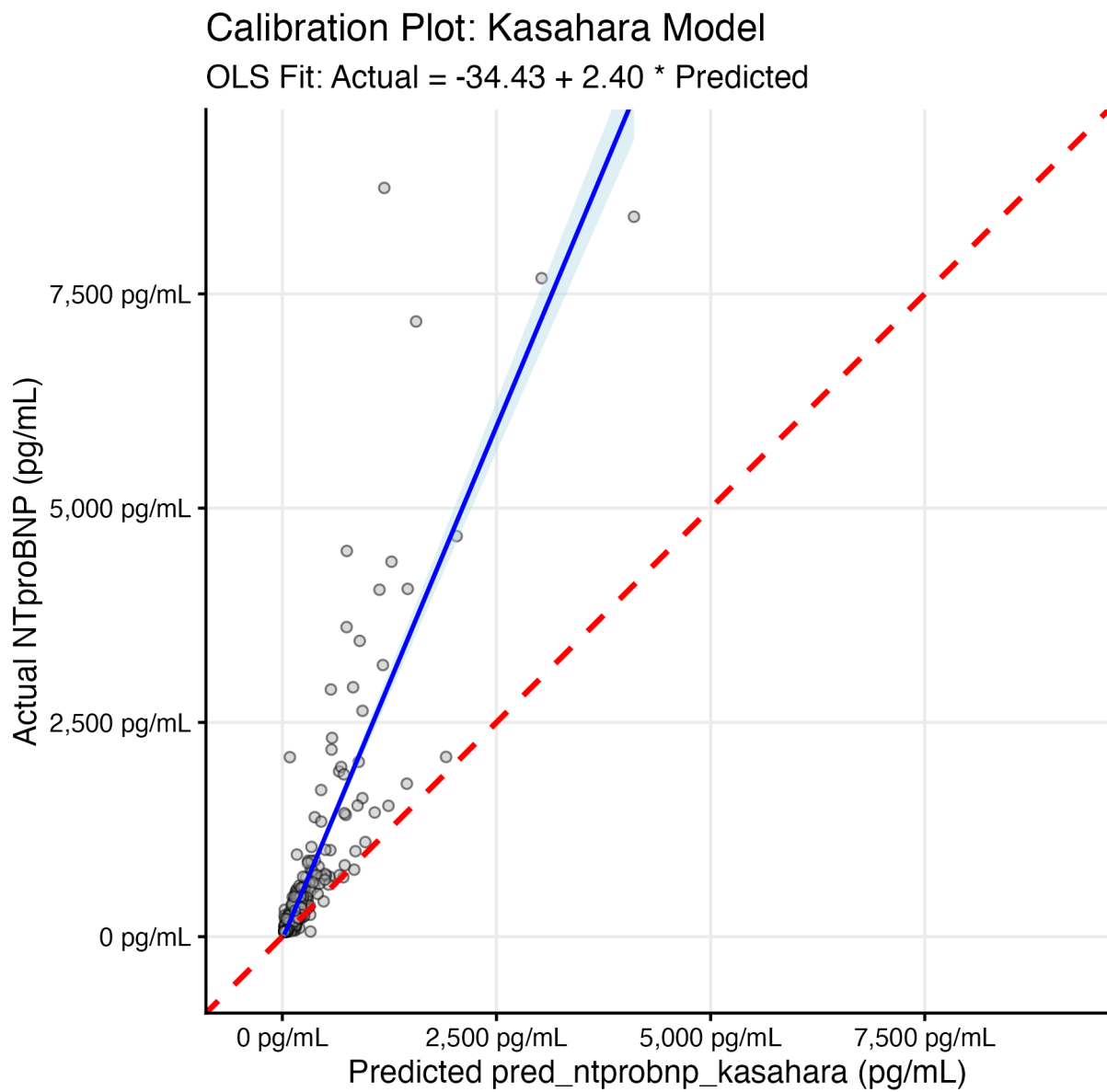


Figure 2: Kasahara model calibration plot (linear scale).

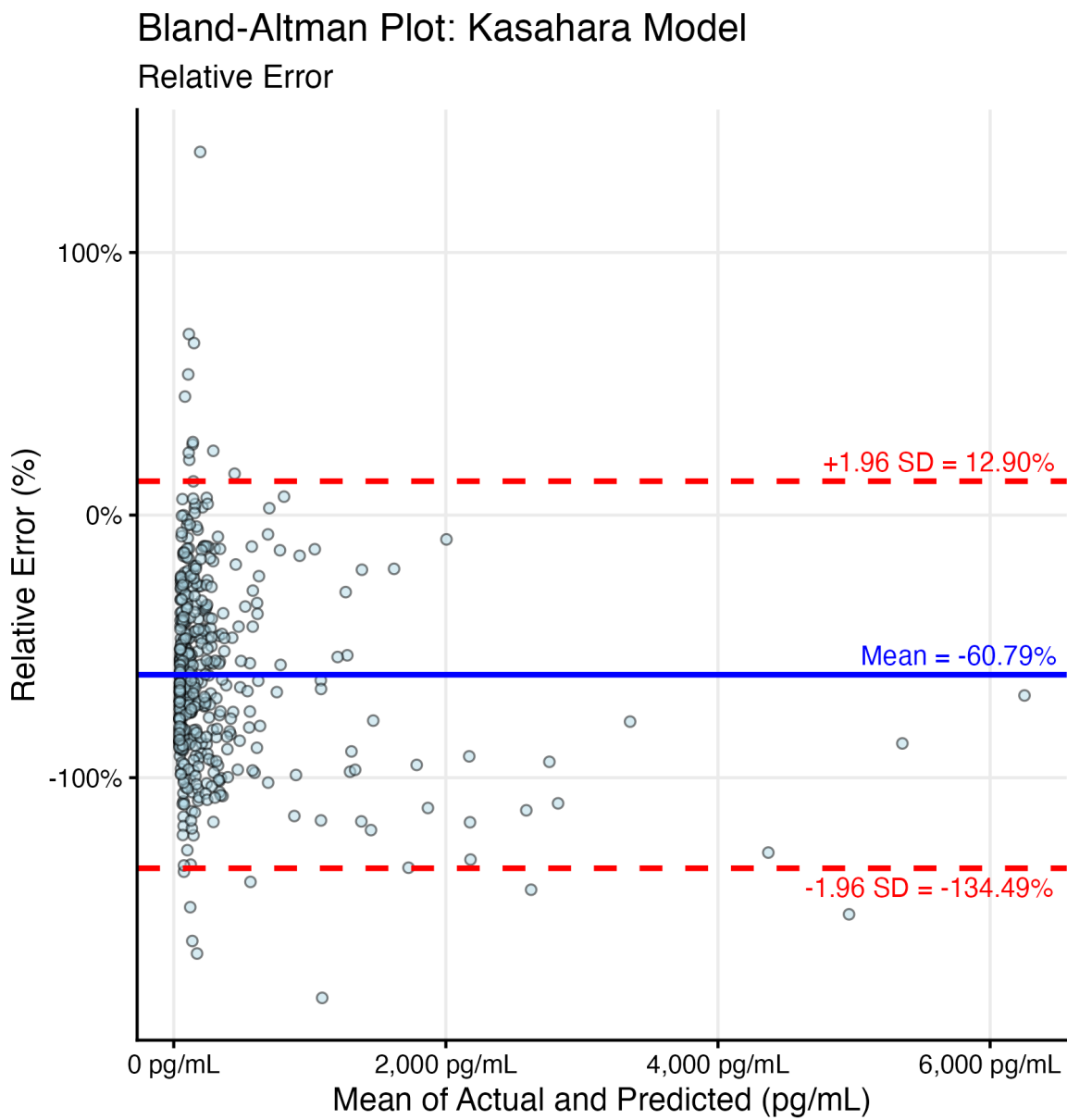


Figure 3: Kasahara model Bland-Altman plot (relative error).

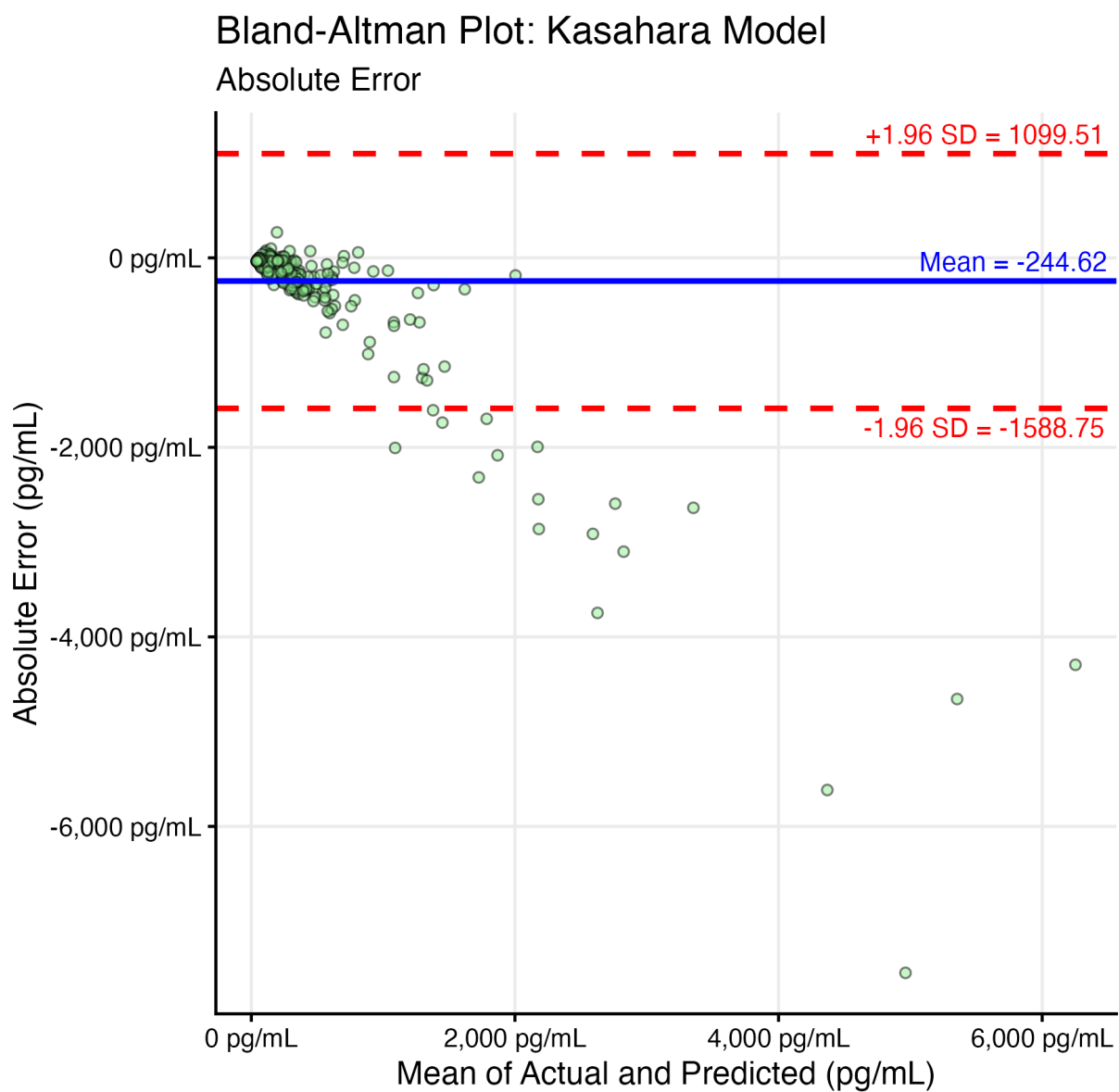
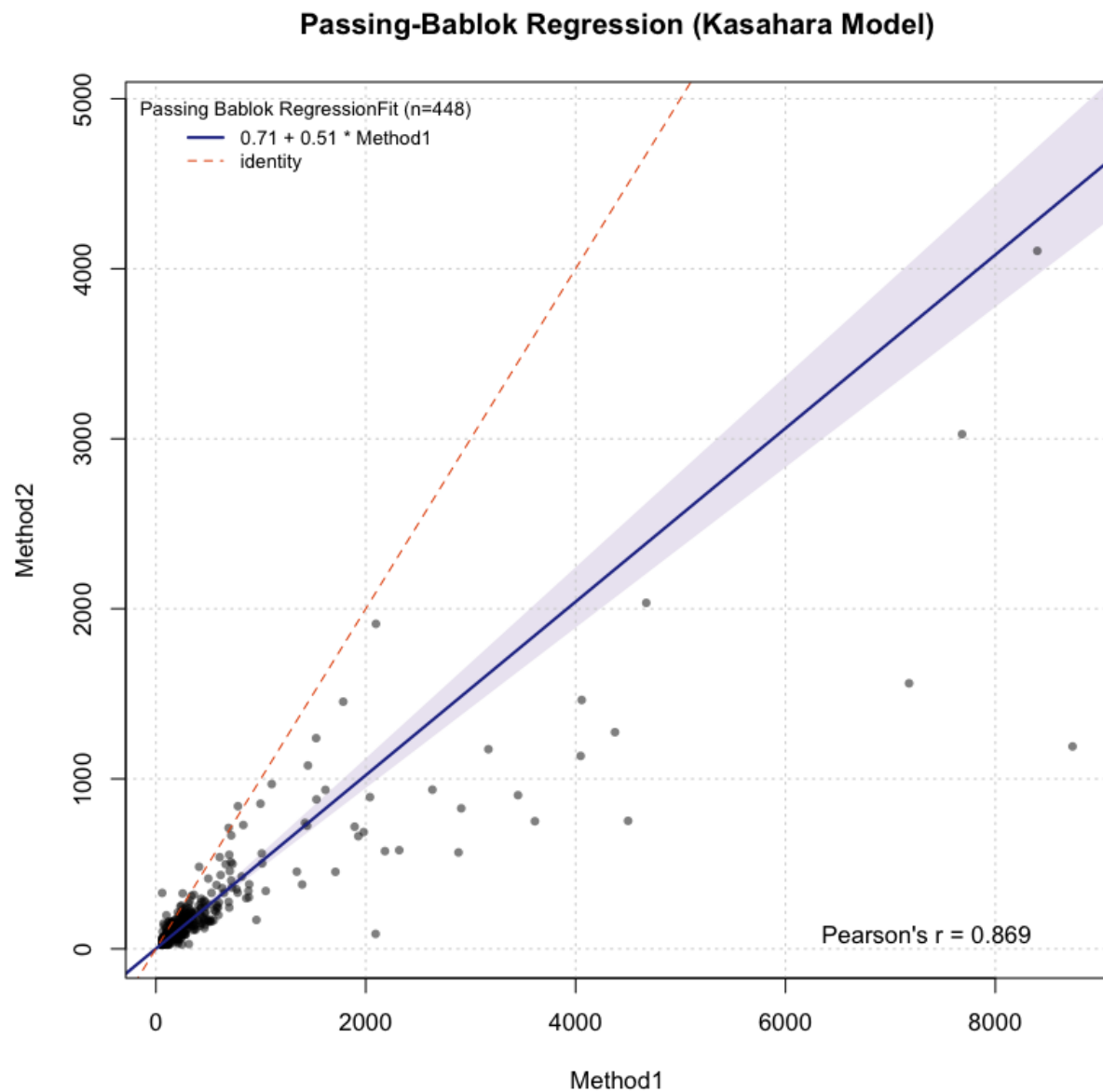


Figure 4: Kasahara model Bland-Altman plot (absolute error).



The 0.95-confidence bounds are calculated with the bootstrap(quantile) method.

Figure 5: Passing-Bablok regression for Kasahara model (measured vs predicted).

4 Updated Conversion Formula (Recalibrated Model)

A new model was fit on log-transformed NT-proBNP with the specified predictors and interactions. Passing-Bablok regression (bootstrap) was used for recalibration.

Bootstrapped Passing-Bablok summary not found.

Table 3: Apparent performance metrics for the recalibrated model (median prediction, PaBa).

metric	estimator	estimate
rmse	standard	406.516
rsq	standard	0.826
mae	standard	145.931
pearson_r	pearson	0.909
accuracy	binary	0.902
sens	binary	0.825
spec	binary	0.936
ppv	binary	0.850
npv	binary	0.924
kap	binary	0.767
accuracy	multiclass	0.757
sens	macro	0.744
spec	macro	0.913
ppv	macro	0.761
npv	macro	0.913
kap	multiclass	0.649
weighted_kappa_quadratic	weighted	0.835
calibration_intercept	OLS	41.579
calibration_slope	OLS	0.946
paba_intercept	PaBa	0.603
paba_intercept_lower	PaBa	-4.653
paba_intercept_upper	PaBa	5.641
paba_slope	PaBa	0.993
paba_slope_lower	PaBa	0.926
paba_slope_upper	PaBa	1.068

Table 4: Optimism-corrected performance metrics (bootstrap).

metric	estimator	apparent estimate	mean optimism	optimism corrected	lower ci 95	upper ci 95	n boots
accuracy	binary	0.902	-0.011	0.913	0.885	0.913	10
accuracy	multiclass	0.757	0.022	0.735	0.684	0.773	10
calibration_intercept	OLS	41.579	-36.194	77.773	-78.263	309.166	10
calibration_slope	OLS	0.946	0.066	0.880	0.195	1.532	10
kap	binary	0.767	-0.026	0.793	0.732	0.798	10
kap	multiclass	0.649	0.028	0.621	0.555	0.671	10
mae	standard	145.931	-95.435	241.366	158.292	362.648	10
npv	binary	0.924	-0.015	0.939	0.901	0.958	10
npv	macro	0.913	0.007	0.906	0.889	0.919	10
paba_intercept	PaBa	0.603	0.408	0.195	-9.923	12.800	10
paba_intercept_lower	PaBa	-4.653	8.333	-12.986	-33.317	1.040	10
paba_intercept_upper	PaBa	5.641	-4.919	10.560	2.620	20.909	10
paba_slope	PaBa	0.993	-0.010	1.003	0.854	1.106	10
paba_slope_lower	PaBa	0.926	0.058	0.868	0.755	0.962	10
paba_slope_upper	PaBa	1.068	-0.124	1.191	0.993	1.404	10
pearson_r	pearson	0.909	0.119	0.790	0.477	0.887	10
ppv	binary	0.850	-0.003	0.852	0.804	0.873	10
ppv	macro	0.761	0.037	0.724	0.656	0.773	10
rmse	standard	406.516	-507.175	913.691	425.049	2511.130	10
rsq	standard	0.826	0.186	0.641	0.257	0.787	10
sens	binary	0.825	-0.033	0.857	0.779	0.901	10
sens	macro	0.744	0.018	0.726	0.674	0.785	10
spec	binary	0.936	-0.001	0.937	0.912	0.947	10
spec	macro	0.913	0.007	0.906	0.890	0.921	10
weighted_kappa_quadratic	weighted	0.835	-0.002	0.837	0.804	0.850	10

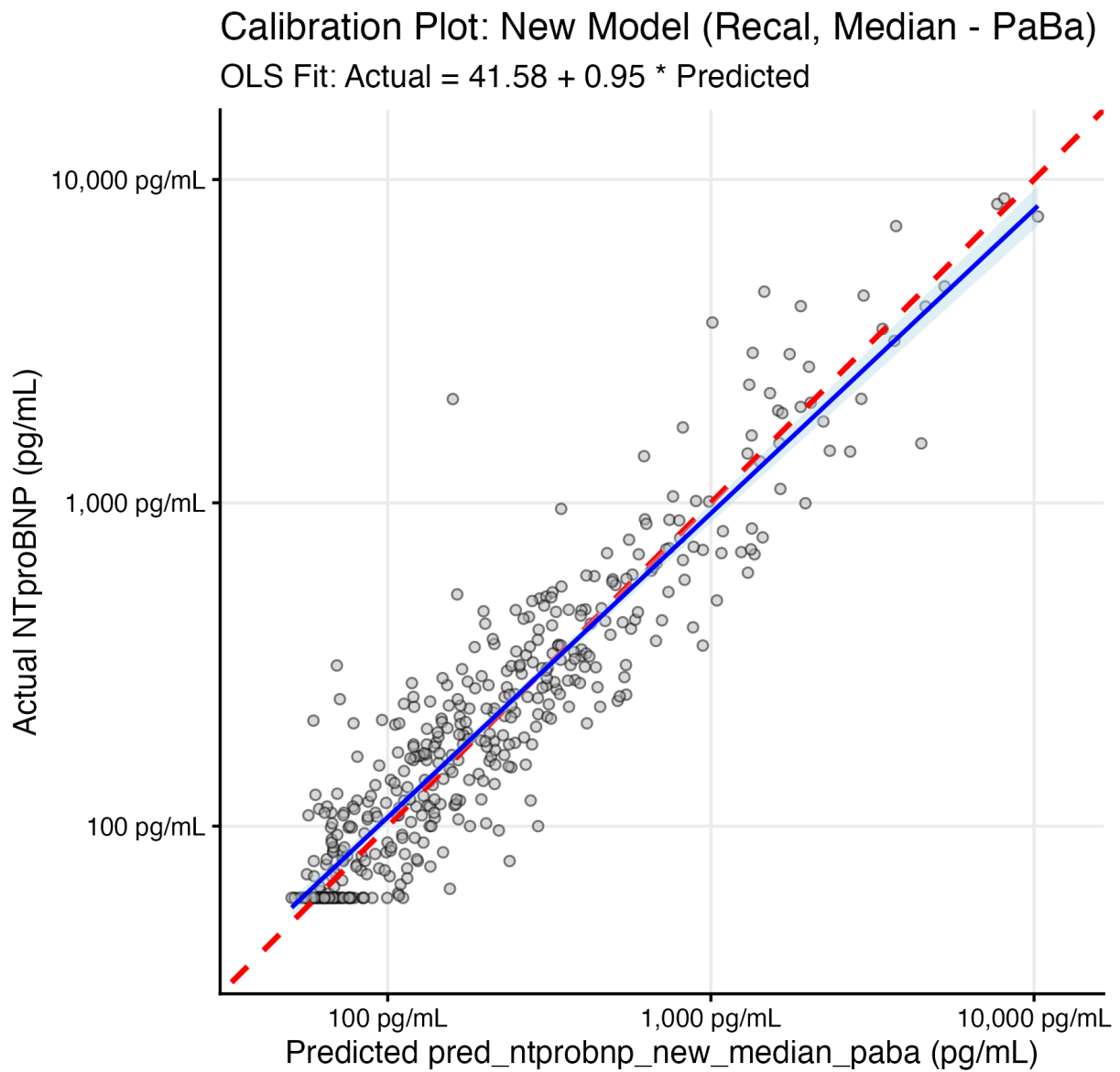


Figure 6: Recalibrated model calibration plot (log scale, median PaBa).

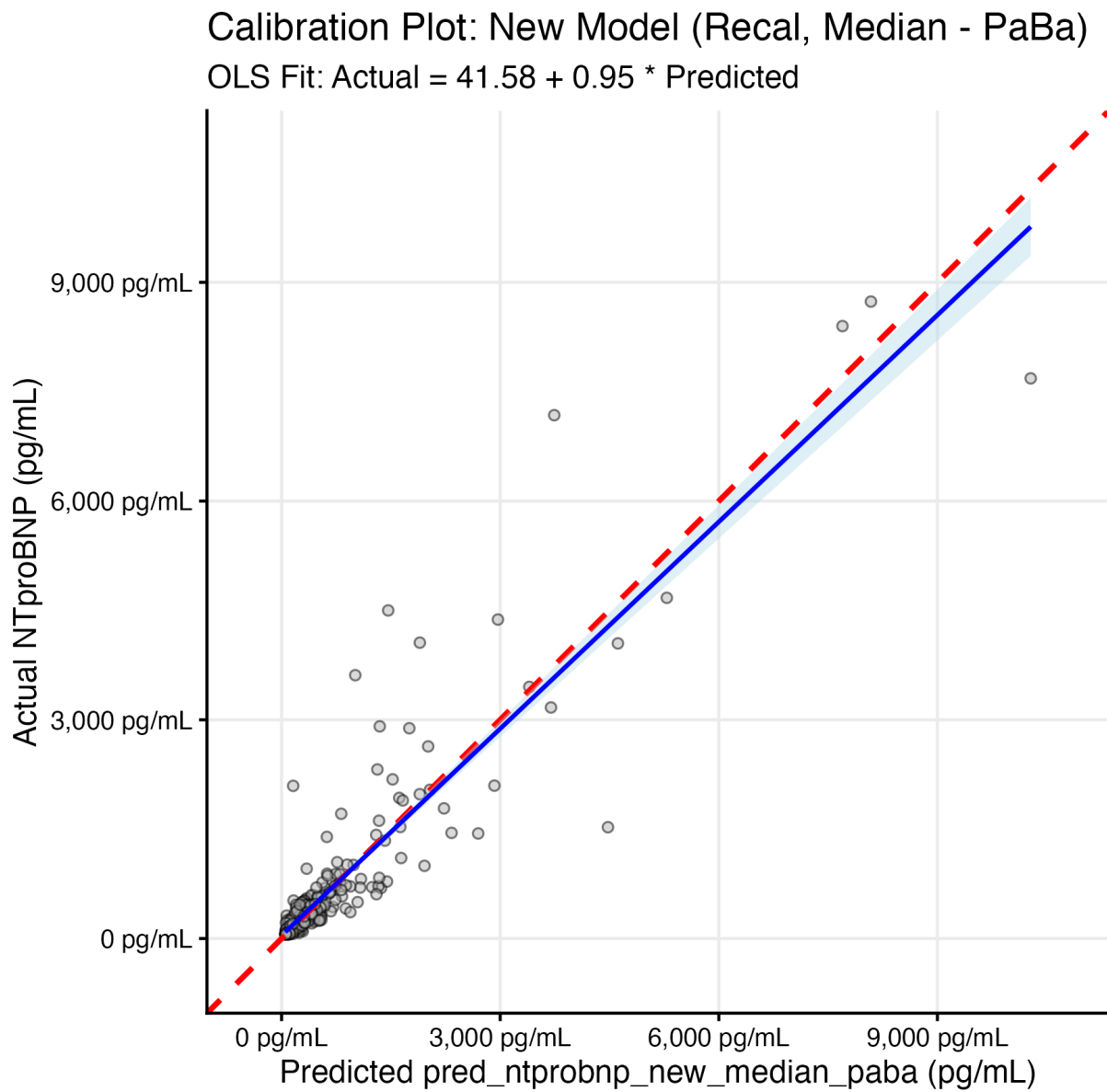


Figure 7: Recalibrated model calibration plot (linear scale, median PaBa).

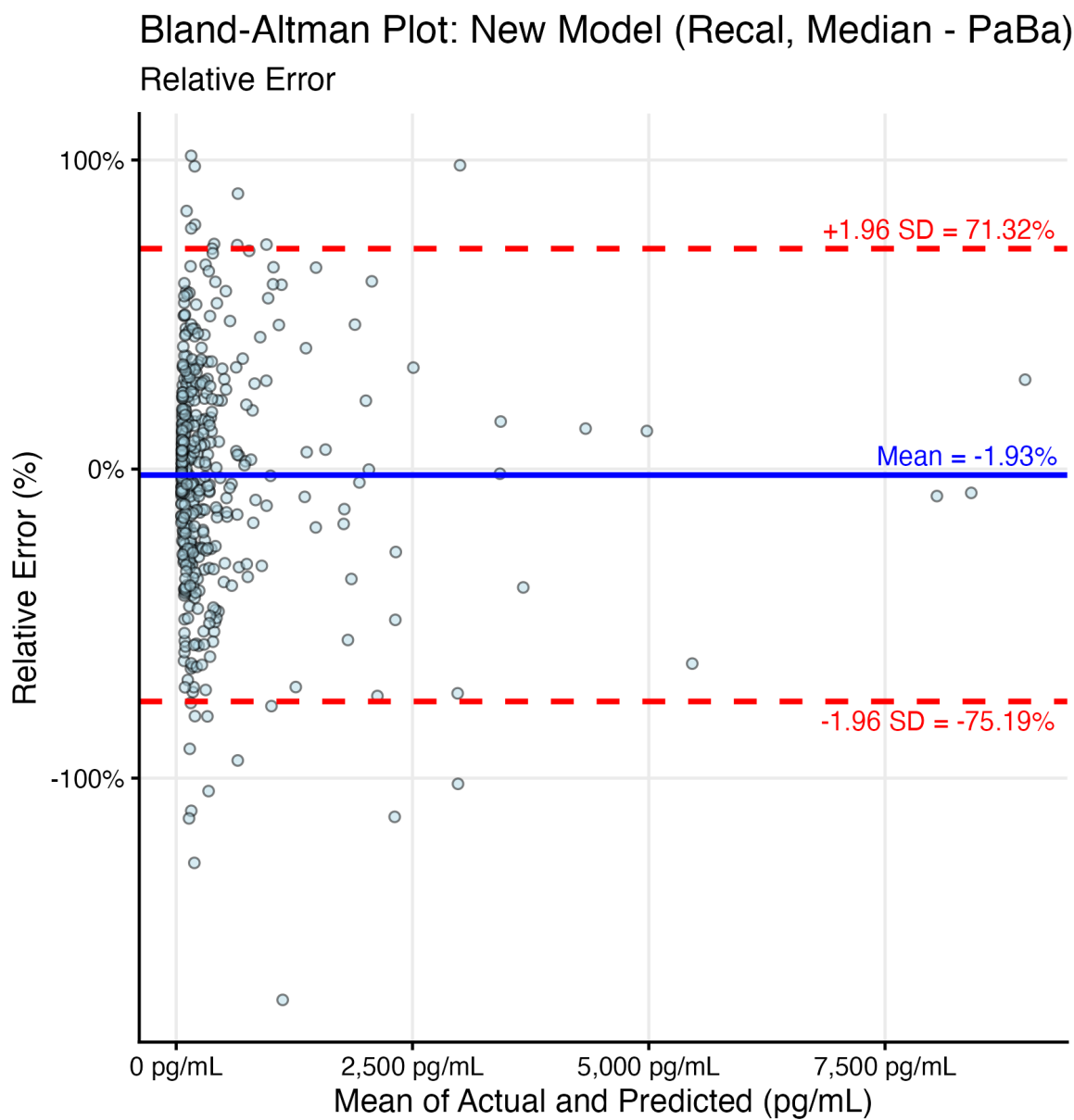


Figure 8: Recalibrated model Bland-Altman plot (relative error, median PaBa).

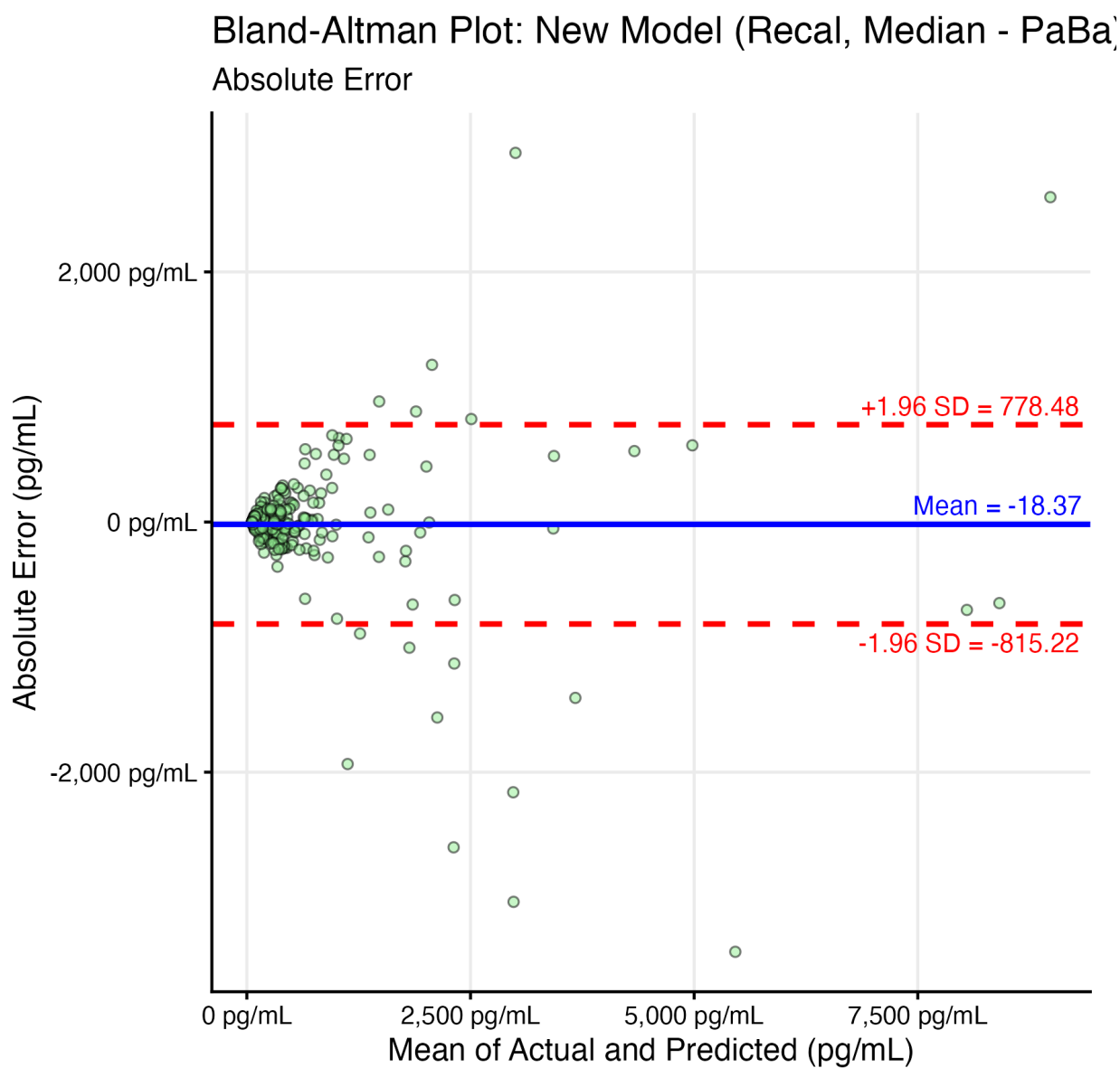
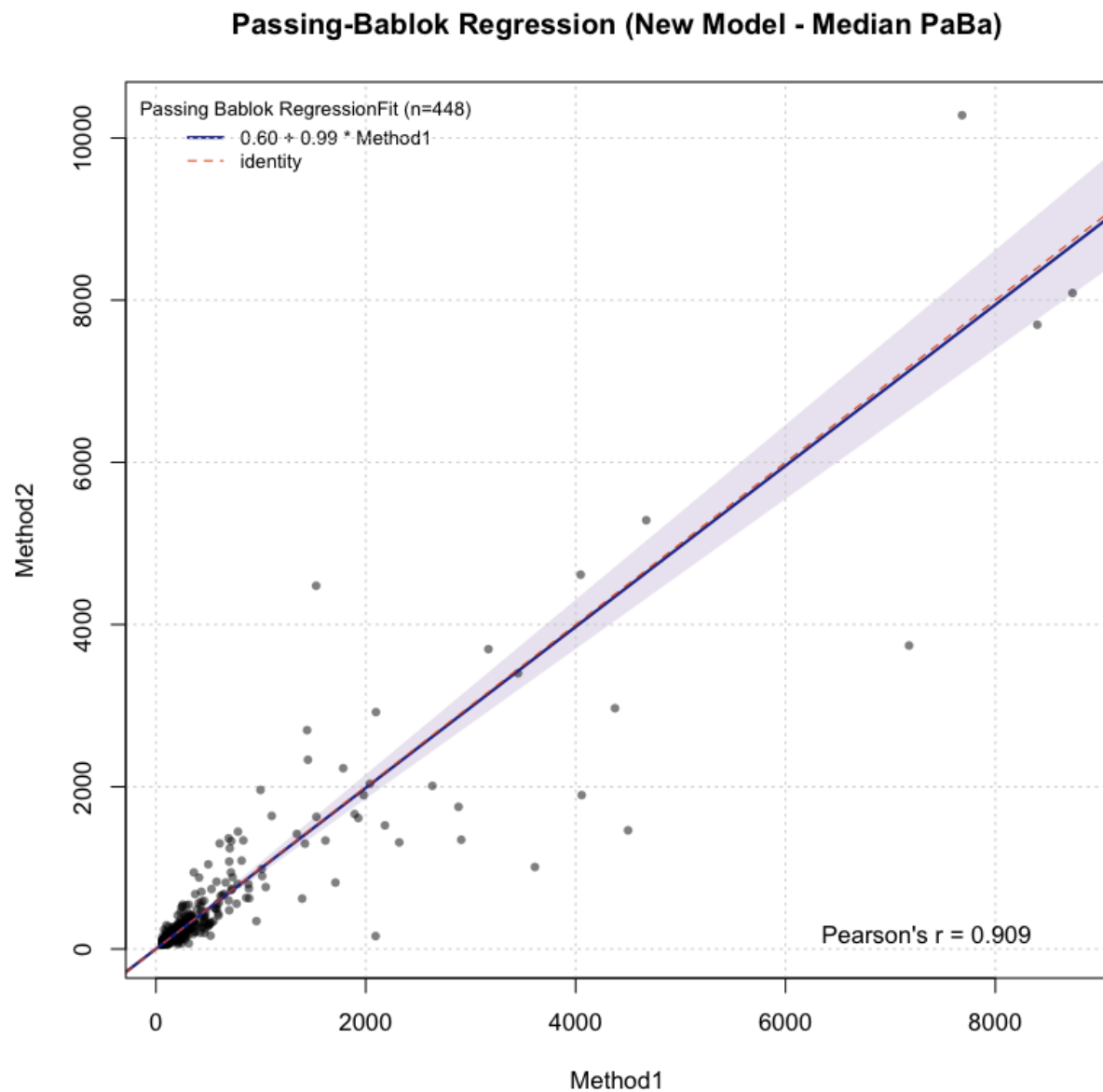


Figure 9: Recalibrated model Bland-Altman plot (absolute error, median PaBa).



The 0.95-confidence bounds are calculated with the bootstrap(quantile) method.

Figure 10: Passing-Bablok regression for recalibrated model (median PaBa, measured vs predicted).

5 Secondary Analysis: Outcome Associations

Logistic regression models evaluate associations between biomarker categories and the 30-day composite outcome.

Table 5: Association between measured NT-proBNP categories and the composite outcome (odds ratios).

part	term	statistic	GLM	Firth GLM
estimates	(Intercept)	estimate	0.047 [0.021, 0.081]	0.055 [0.032, 0.096]
estimates	(Intercept)	p.value	(<0.001)	(<0.001)
estimates	NTproBNP_cat.L	estimate	14.203 [3.992, 105.672]	10.992 [3.106, 38.894]
estimates	NTproBNP_cat.L	p.value	(<0.001)	(<0.001)
estimates	NTproBNP_cat.Q	estimate	1.141 [0.234, 3.703]	1.300 [0.430, 3.934]
estimates	NTproBNP_cat.Q	p.value	(0.839)	(0.642)
estimates	NTproBNP_cat.C	estimate	1.342 [0.447, 3.677]	1.331 [0.528, 3.357]
estimates	NTproBNP_cat.C	p.value	(0.567)	(0.544)

Table 6: Association between derived BNP categories and the composite outcome (odds ratios).

part	term	statistic	GLM	Firth GLM
estimates	(Intercept)	estimate	0.057 [0.025, 0.104]	0.067 [0.037, 0.120]
estimates	(Intercept)	p.value	(<0.001)	(<0.001)
estimates	BNP_cat_derived.L	estimate	18.708 [4.174, 150.813]	14.769 [3.605, 60.501]
estimates	BNP_cat_derived.L	p.value	(<0.001)	(<0.001)
estimates	BNP_cat_derived.Q	estimate	1.380 [0.269, 4.698]	1.622 [0.502, 5.240]
estimates	BNP_cat_derived.Q	p.value	(0.636)	(0.419)
estimates	BNP_cat_derived.C	estimate	1.214 [0.450, 3.195]	1.188 [0.497, 2.843]
estimates	BNP_cat_derived.C	p.value	(0.687)	(0.698)

Table 7: Association between predicted NT-proBNP categories (median PaBa) and the composite outcome (odds ratios).

part	term	statistic	GLM	Firth GLM
estimates	(Intercept)	estimate	0.048 [0.021, 0.083]	0.056 [0.032, 0.098]
estimates	(Intercept)	p.value	(<0.001)	(<0.001)
estimates	pred_new_median_cat.L	estimate	16.130 [4.516, 120.232]	12.456 [3.507, 44.235]
estimates	pred_new_median_cat.L	p.value	(<0.001)	(<0.001)
estimates	pred_new_median_cat.Q	estimate	1.137 [0.233, 3.700]	1.294 [0.426, 3.925]
estimates	pred_new_median_cat.Q	p.value	(0.843)	(0.649)
estimates	pred_new_median_cat.C	estimate	1.369 [0.456, 3.754]	1.357 [0.538, 3.426]
estimates	pred_new_median_cat.C	p.value	(0.541)	(0.518)

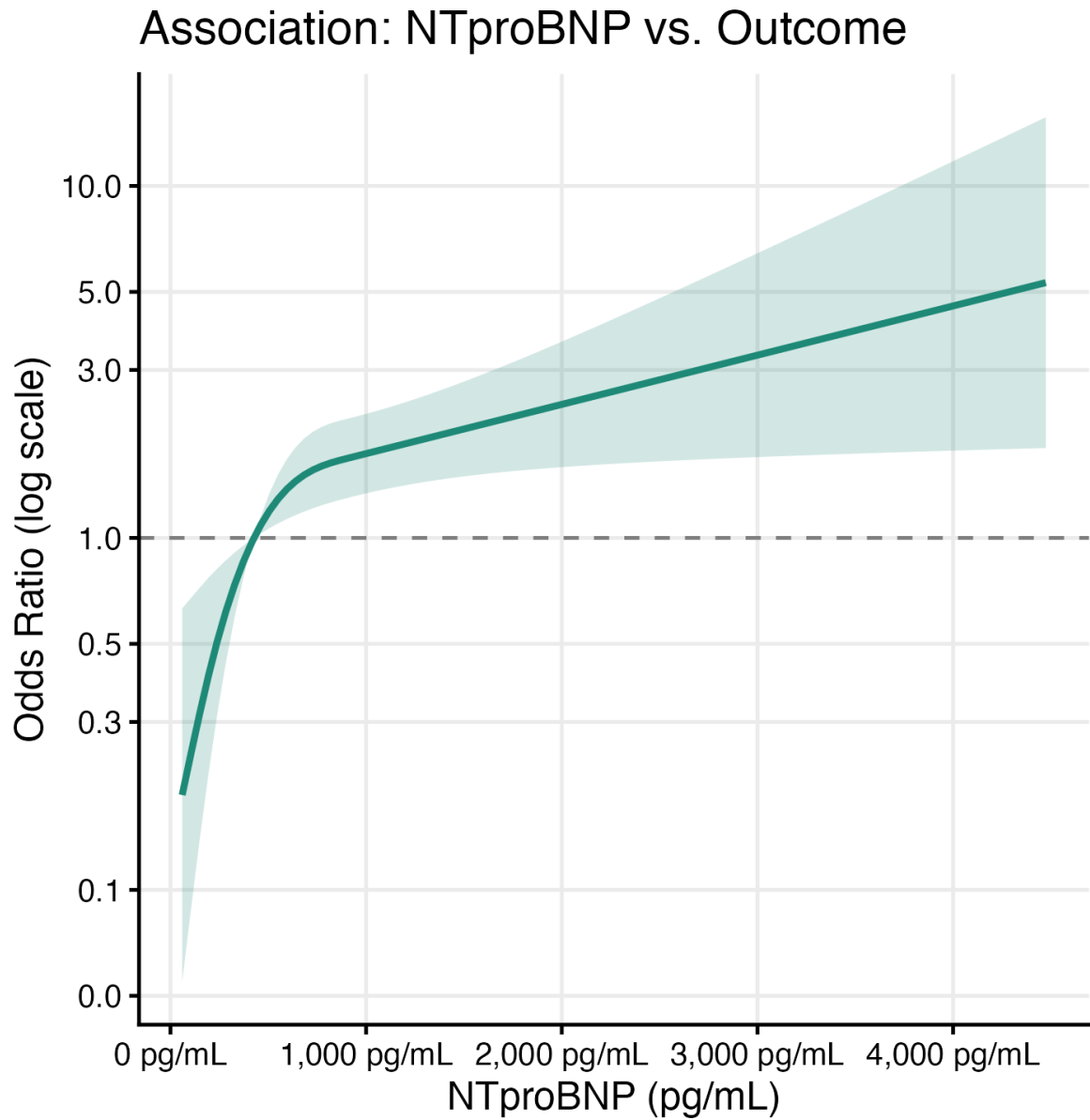


Figure 11: Association of measured NT-proBNP with outcome (spline model).

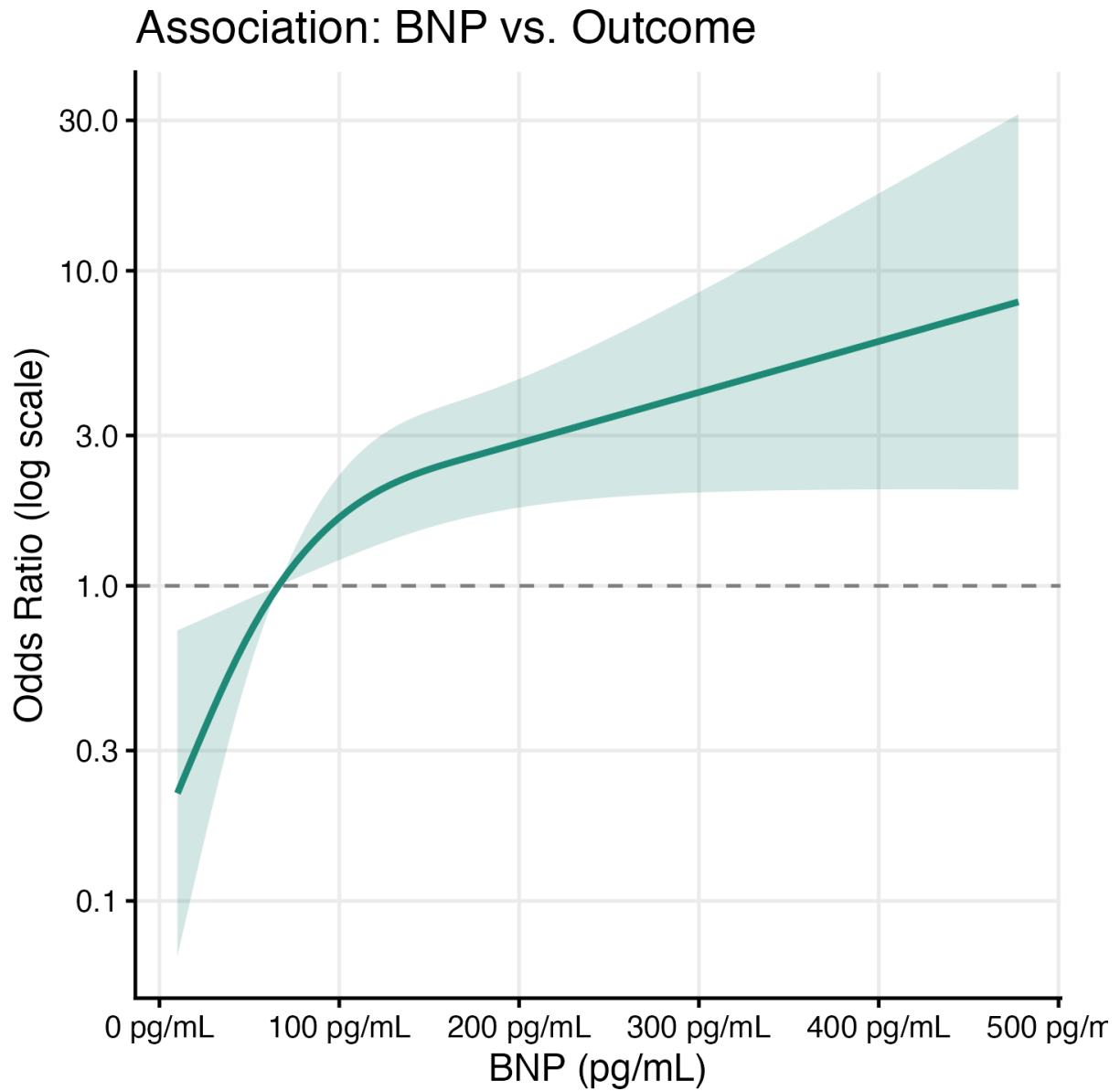


Figure 12: Association of measured BNP with outcome (spline model).

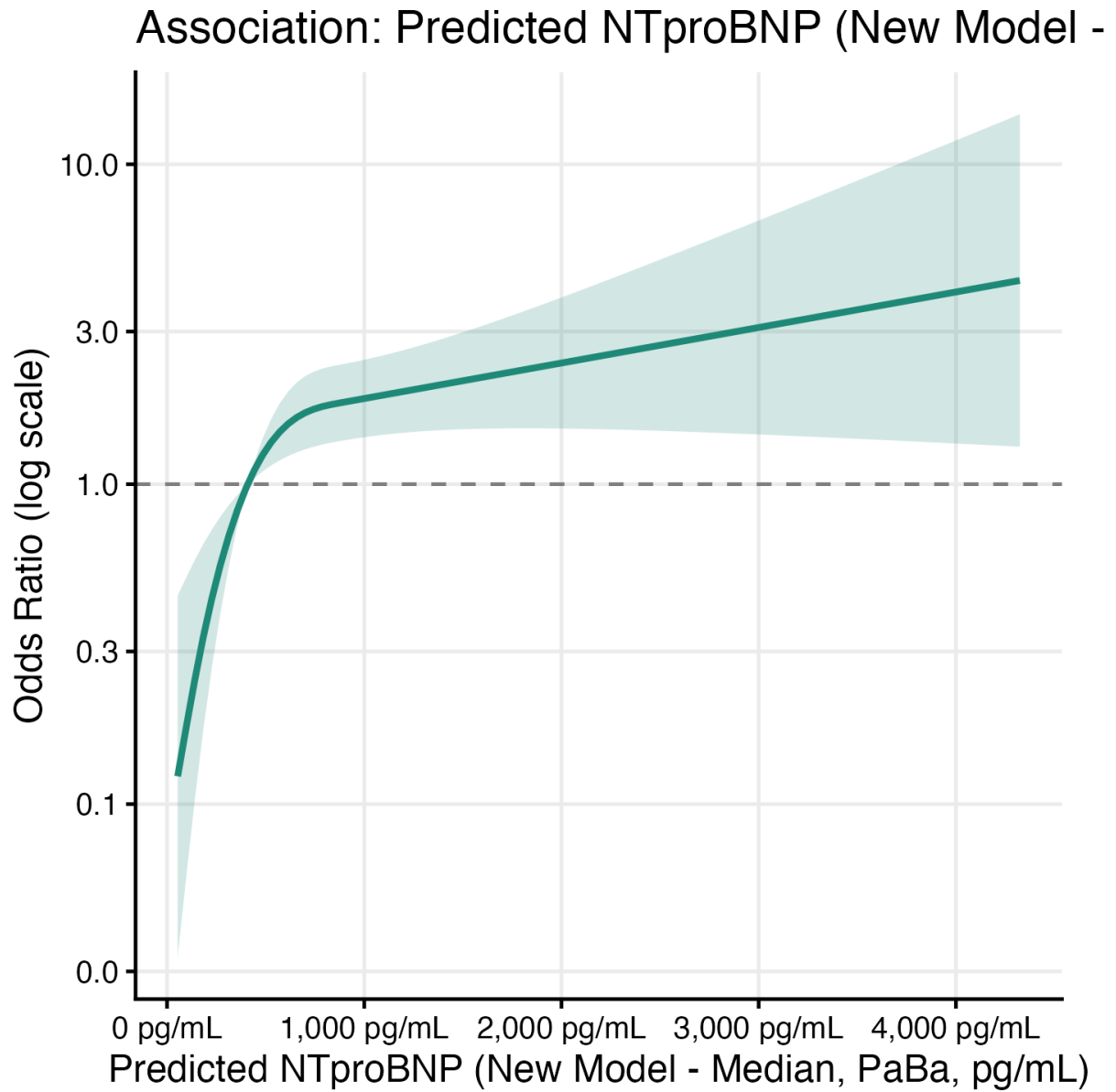


Figure 13: Association of predicted NT-proBNP (new model, median PaBa) with outcome (spline model).